WORKED SOLUTION

STEP III 1994, Question 5

Pure • Maclaurin Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q5

The key idea

Differentiating $\sin^{-1} x$ once gives a square root, and differentiating again gives a relation between f'' and f' that has no square roots in it. Leibniz's theorem then turns that one equation into a recurrence for all the higher derivatives. At $x = 0$ most of the terms vanish, which gives a simple recurrence for the Maclaurin coefficients. For g, the laws of logarithms reduce everything to the familiar series for $\ln(1 \pm x)$.

Solution

The differential equation

For $-1 < x < 1$,

$$\mathrm{f}'(x) = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}, \qquad \mathrm{f}''(x) = -\tfrac{1}{2}(1-x^2)^{-3/2} \cdot (-2x) = \frac{x}{(1-x^2)^{3/2}}.$$

Multiplying the second by $1 - x^2$ gives $(1-x^2)\mathrm{f}''(x) = x(1-x^2)^{-1/2} = x\mathrm{f}'(x)$, so

$$(1-x^2)\mathrm{f}''(x) - x\mathrm{f}'(x) = 0,$$

as required.

The recurrence

Differentiate this identity n times using Leibniz's theorem, $(uv)^{(n)} = \sum_r \binom{n}{r} u^{(r)} v^{(n-r)}$. In the first product we take $u = 1 - x^2$. Its derivatives are $-2x$, -2 and then 0, so only three terms survive:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[(1-x^2)\mathrm{f}''\right] = (1-x^2)\mathrm{f}^{(n+2)} + n(-2x)\mathrm{f}^{(n+1)} + \binom{n}{2}(-2)\mathrm{f}^{(n)}.$$

In the second product $u = x$, and only two terms survive:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[x\mathrm{f}'\right] = x\mathrm{f}^{(n+1)} + n\mathrm{f}^{(n)}.$$

Subtracting, and using $2\binom{n}{2} = n(n-1)$,

$$\begin{aligned} 0 &= (1-x^2)\mathrm{f}^{(n+2)} - (2n+1)x\mathrm{f}^{(n+1)} - (n(n-1)+n)\mathrm{f}^{(n)} \\ &= (1-x^2)\mathrm{f}^{(n+2)} - (2n+1)x\mathrm{f}^{(n+1)} - n^2\mathrm{f}^{(n)}, \end{aligned}$$

which is the required result. It holds for every $n \geq 1$, and for $n = 0$ it is just the first equation. (If you prefer, you can prove it by induction instead: differentiate the case n once and collect terms to get the case $n+1$. That is the same calculation done one step at a time.)

The Maclaurin series of f

? **Ask yourself** Which value of x makes this recurrence easy to use?

Put $x = 0$. The middle term vanishes and $1 - x^2 = 1$, so

$$\mathrm{f}^{(n+2)}(0) = n^2\,\mathrm{f}^{(n)}(0) \qquad (n \geq 0).$$

We start from $\mathrm{f}(0) = \sin^{-1} 0 = 0$ and $\mathrm{f}'(0) = 1$.

Even derivatives. Since $\mathrm{f}(0) = 0$, the recurrence makes every even derivative zero at 0. This is what we expect, because $\sin^{-1}$ is an odd function.

Odd derivatives. With $n = 1, 3, 5, \ldots$,

$$\mathrm{f}'''(0) = 1^2, \quad \mathrm{f}^{(5)}(0) = 3^2 \cdot 1^2, \quad \mathrm{f}^{(7)}(0) = 5^2 \cdot 3^2 \cdot 1^2,$$

and in general

$$\mathrm{f}^{(2k+1)}(0) = (1 \cdot 3 \cdot 5 \cdots (2k-1))^2 \qquad (k \geq 1).$$

Therefore

$$\sin^{-1} x = x + \frac{1^2}{3!}x^3 + \frac{1^2 \cdot 3^2}{5!}x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!}x^7 + \cdots = x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + \cdots.$$

The coefficient of x^{2k+1} simplifies neatly. Write $(2k+1)! = (1 \cdot 3 \cdots (2k-1))(2 \cdot 4 \cdots 2k)(2k+1)$ and cancel one copy of the odd product:

$$a_{2k+1} = \frac{(1 \cdot 3 \cdots (2k-1))^2}{(2k+1)!} = \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots (2k)} \cdot \frac{1}{2k+1}.$$

The ratio test shows that this series converges for $|x| < 1$, since $a_{2k+3}/a_{2k+1} \to 1$. That it actually equals $\sin^{-1} x$ there can be confirmed by integrating the binomial series for $(1-x^2)^{-1/2}$ term by term, which gives exactly the same coefficients.

The function g

By the laws of logarithms, $\mathrm{g}(x) = \frac{1}{2}(\ln(1+x) - \ln(1-x))$. For $|x| < 1$,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \qquad \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots.$$

In the difference the even powers cancel and the odd powers double. So

$$g(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}.$$

(You may recognise g as $\tanh^{-1} x$.)

Comparing coefficients

? **Ask yourself** Both coefficients have a factor $\frac{1}{2k+1}$. What is left to compare?

For $n \geq 1$, the coefficient of x^{2n+1} is $\dfrac{1}{2n+1}$ in g and

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} \cdot \frac{1}{2n+1}$$

in f. Each of the n factors $\frac{2r-1}{2r}$ lies strictly between 0 and 1, and there is at least one of them because $n \geq 1$. So their product is strictly less than 1, and hence

$$\frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} \cdot \frac{1}{2n+1} < \frac{1}{2n+1}.$$

The coefficient in g is therefore greater, as required. When $n = 0$ both coefficients equal 1, which is why the question excludes that case.

Checks

- At $x = -\frac{1}{2}$, the series summed up to x^{59} gives -0.5235987756, matching $\sin^{-1}(-\frac{1}{2}) = -\pi/6$.
- The coefficients $1, \frac{1}{6}, \frac{3}{40}, \frac{5}{112}$ agree with those generated directly from $f^{(n+2)}(0) = n^2 f^{(n)}(0)$. The comparison $\frac{1}{6} < \frac{1}{3}$, $\frac{3}{40} < \frac{1}{5}$, $\frac{5}{112} < \frac{1}{7}$ holds as claimed.
- Since every coefficient of f is at most the matching one of g, we should have $\sin^{-1} x < \tanh^{-1} x$ for $0 < x < 1$. At $x = 0.5$: $0.5236 < 0.5493$.

Common slips

- Forgetting the $\binom{n}{2}$ term in Leibniz's theorem, which gives n instead of n^2 in the final coefficient.
- Writing $f^{(2k+1)}(0) = 1 \cdot 3 \cdots (2k-1)$ without the square. Each step of the recurrence multiplies by n^2, not n.
- Dividing by $(2k)!$ or $(2k+1)$ alone instead of $(2k+1)!$ when forming the Maclaurin coefficient.

Beyond STEP

The ratio $\frac{1\cdot3\cdots(2n-1)}{2\cdot4\cdots(2n)}$ equals $\binom{2n}{n}/4^n$, and it behaves like $1/\sqrt{\pi n}$ for large n. This is a consequence of Wallis's product and Stirling's formula. The series for $\sin^{-1} x$ converges at $x = 1$ too, giving $\pi/2$, but proving that needs Abel's theorem on power series at the edge of their interval of convergence. When a function's derivatives satisfy a differential equation, a recurrence for the coefficients follows; this is the method of series solutions for ODEs, covered in Boyce and DiPrima, *Elementary Differential Equations*. For careful proofs of when a Taylor series really equals its function, see Abbott, *Understanding Analysis*.

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